Maths.

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1. necessary condition for a point x^* to be a local minimum is that the gradient of the objective function vanishes at that point.

$$\nabla F(x^*) = 0$$

$\downarrow$

called the first order necessary condition

2. $\nabla F(x) = 0$ means a critical point $\rightarrow$ could be min, max or saddle.

3. $F(x) = x^2 \rightarrow \nabla F(0) = 0$

$F''(x) = 2 > 0$

$\therefore x = 0$ is a min.

4. False, zero gradient $\neq$ always global min.

5. Second order necessary

$\nabla F(x) = 0$ and Hessian is positive semi definite.

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6. $f(x, y) = x^2 + y^2$

$\nabla f(0,0) = 0$

Hessian = [[2,0], [0,2]]

PD = (0,0)

is a strict local min.

7. Necessary → must hold at min
sufficient → guarantees min.

8. KKT conditions:
1. Stationarity
2. Primal feasibility
3. Dual feasibility
4. Complementary slackness

9. In convex problems,
satisfying KKT ⇒ global min.

10. min x^2 s.t. $1 - x \leq 0$

KKT point $x = 1$ $\lambda = 2$

min value = 1

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11. semi definite hessian $\Rightarrow$
local min possible
not strict min

12. Taylor ~~exposition~~ expansion
$\Rightarrow$ if $\nabla F = 0$ and
local min $\Rightarrow v^T \nabla^2 f(x) v \geq 0$

13. KKT helps check optimality
under constraints $\rightarrow$ feasibility
+ cost optimality